

## Acne and Skin Clearing Misinformation on TikTok: Skinfluencers, Transformation Videos, and Unverified Treatment Claims

Goal – To collect and analyse TikTok video content related to acne treatment and skin clearing, (a) assess unverified or clinically inaccurate claims made by skinfluencers and non-medical creators, (b) quantify the before/after transformation video, (c) whether acne transformation videos promoting unverified remedies — achieves higher engagement than clinically accurate dermatologist-led content.

There are few published studies on this topic, but every existing study uses manual coding of 50–200 videos. None has applied NLP or computer vision to classify before/after transformation videos at scale, none has automated skin state classification using CLIP or dermatological image analysis, and none has connected before/after visual claims to treatment accuracy ratings or engagement prediction models.

Example of hashtags - #acne #acnetreatment #acnetransformation #acneclearingskin

Scraping: Apify TikTok Scraper (~\$5/run) for metadata collection across all hashtags above. TikTok Research API for longitudinal creator tracking. pyktok ([github.com/dfreelon/pyktok](https://github.com/dfreelon/pyktok)) for mp4 video download. yt-dlp for caption and subtitle extraction. Whisper large-v3 for audio-to-text transcription. Target: qpprox 4,000–6,000 mp4 downloads prioritising transformation-format videos (week-by-week clearing series) and videos with before/after text in title or caption. Date range: January 2024 – December 2025.

The following example taxonomy covers the most prevalent unverified or clinically inaccurate claims in TikTok acne content. Clinical verdicts are drawn from AAD (American Academy of Dermatology) guidelines and published evidence from JAMA Dermatology and the British Journal of Dermatology. This taxonomy can be the classification framework for the NLP and LLM-based accuracy scoring pipeline.

| Claim Category                   | Example TikTok Claim (paraphrased)                                                         | Clinical Verdict (AAD)                                                                                               | Before/After Video Potential                                                                      |
|----------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Dietary cure — dairy elimination | 'I cut out dairy for 30 days and my cystic acne completely cleared — no medication needed' | PARTIALLY SUPPORTED for some patients; no RCT evidence for universal dairy elimination; AAD calls evidence 'limited' | VERY HIGH — 30-day before/after series; highly shareable; confirmation bias drives positive posts |

|                                             |                                                                                                       |                                                                                                                                                                         |                                                                                                           |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Dietary cure — sugar/gluten                 | 'Going sugar-free cured my hormonal acne in 3 weeks — this is what your dermatologist won't tell you' | UNSUPPORTED — high-glycaemic diet is associated with acne but elimination is not clinically proven as primary treatment                                                 | HIGH — week-by-week skin transformation with dietary protocol attribution                                 |
| Gut health / microbiome cure                | 'Fixing your gut with probiotics and cutting seed oils cleared my acne from the inside out'           | UNSUPPORTED as primary treatment — gut-skin axis research is early-stage; no validated protocol                                                                         | HIGH — supplement promotion + before/after; functional medicine framing adds pseudoscientific credibility |
| DIY topicals — toothpaste/garlic            | 'Put toothpaste on your pimple overnight and it's gone by morning — works every time'                 | FALSE AND HARMFUL — toothpaste causes chemical burns and contact dermatitis; garlic causes skin necrosis                                                                | MODERATE — single-pimple before/after overnight; very high short-term engagement                          |
| DIY chemical peels at unsafe concentrations | 'I use 30% glycolic acid at home — my skin has never been clearer'                                    | DANGEROUS — high-concentration AHAs/BHAs require clinical supervision; risk of chemical burns, PIH, scarring                                                            | HIGH — skin texture before/after; influencer tutorial format; product link in bio                         |
| Anti-isotretinoin framing                   | 'Accutane destroyed my mental health and ruined my gut — try this natural alternative instead'        | MISLEADING — isotretinoin is the most effective acne treatment available; mental health link is contested and context-dependent; discouraging use is clinically harmful | HIGH — personal testimony + before/after alternative treatment story; emotionally compelling              |
| Hormone balancing supplements               | 'DIM, spearmint tea, and zinc cleared my hormonal acne without a prescription in 6 weeks'             | UNSUPPORTED as evidence-based primary treatment — limited, low-quality evidence only; AAD                                                                               | VERY HIGH — supplement stack before/after is the dominant hormonal acne misinformation format             |

|                       |                                                                                              |                                                                                                                                                              |                                                                                                                                |
|-----------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
|                       |                                                                                              | does not recommend as first-line                                                                                                                             |                                                                                                                                |
| Ice/cold therapy      | 'Icing your face for 10 minutes reduces inflammation and prevents breakouts — do this daily' | NOT EVIDENCE-BASED — no clinical trials support ice as acne treatment; temporary vasoconstriction only                                                       | LOW — before/after less convincing; more process-oriented content                                                              |
| LED / at-home devices | 'This \$30 red light device cleared my acne completely — dermatologists hate this'           | PARTIALLY SUPPORTED for low-power devices in mild acne only — consumer devices significantly underpowered vs. clinical devices                               | MODERATE — device unboxing + 6-week before/after; high product promotion potential                                             |
| Sunscreen causes acne | 'Sunscreen broke me out — I stopped wearing it and my acne cleared in a week'                | FALSE AND DANGEROUS — non-comedogenic sunscreen is clinically recommended; avoiding sun protection risks post-inflammatory hyperpigmentation and skin cancer | HIGH — before/after narrative easy to construct; removes a daily product which creates visible change through other mechanisms |

Questions for investigation -

What proportion of TikTok acne content uses the before/after transformation format, how does the claimed treatment in transformation videos map onto the misinformation taxonomy above, and what skin improvement is actually visible in frames extracted from the beginning versus end of each video as classified by CLIP and YOLOv8?

Assess profiles of Tiktok creators - are these videos by medical professionals or by common influencers?

What are the key features that are associated with viewer engagement ? -- Do transformation videos containing unverified treatment claims achieve significantly higher engagement (views, likes, shares) than clinically accurate dermatologist content, after controlling for creator follower count, video length, and posting date? Is there a measurable 'transformation premium' — the excess engagement associated with before/after format above and beyond other content types?

Suggested Pipeline: (1) detect before/after format using zero-shot CLIP classification of thumbnails and mid-video frames ('inflamed acne skin', 'clear and smooth skin', 'before and after skin transformation', 'close-up of pimples or lesions'); (2) extract the first 10 seconds and last 10 seconds of each transformation video as representative frames; (3) apply a skin lesion severity scorer using YOLOv8 fine-tuned on the ACNE04 dataset (a publicly available annotated acne image dataset with 4 severity grades); (4) compute delta severity score (start minus end) as an objective measure of visible skin improvement. Test whether delta severity correlates with claimed treatment type — i.e., whether unverified remedy videos show the same magnitude of visual improvement as clinically validated treatment videos.

## References

- Iyengar, Saldanha & Chong (2025). #Acne: A Thematic Qualitative Analysis of Acne Content on TikTok. *Australasian Journal of Dermatology*. [2B views, 11% clinician, 5 themes — primary study] <https://pmc.ncbi.nlm.nih.gov/articles/PMC12062723/>
- Irfan, Yasin & Yaqoob (2023). Navigating Digital Dermatology: An Analysis of Acne-Related Content on TikTok. *Cureus*, 15, e45226. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10576439/>
- Nigro, Osman, Saal & Smith (2025). Information Regarding Dermatology as Seen on the Social Media Platform TikTok. *JMIR Dermatology*, 8, e59597. <https://derma.jmir.org/2025/1/e59597>
- Vidal et al. (2025). Evaluating Quality and Reliability of Most-Viewed TikTok Videos About Spironolactone. *Journal of Drugs in Dermatology* — #spironolactone: 108M views, significant accuracy gap. <https://mediarelations.gwu.edu/study-reveals-risk-misinformation-about-acne-medications-tiktok>
- Asare, Sefa, Haddad & Obeng (2025). Isotretinoin on TikTok: A Qualitative Analysis of Attitudes, Perspectives, and Knowledge Dissemination. *Cureus*, 17, e80244. <https://pubmed.ncbi.nlm.nih.gov/40196056/>
- Bal, Karaosmanoglu, Temel & Orenay (2025). Trust in Dermatologists Versus Social Media Influencers Among Acne Patients. *Cureus*, 17, e83930. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12068905/>
- Dermatology Times (2025). #DermTok: Navigating TikTok's Acne Advice Misinformation. [Only 11% of #acne videos from clinicians; 129B views; 14% product endorsements] <https://www.dermatologytimes.com/view/dermtok-navigating-tiktok-s-acne-advice-misinformation>
- Yousaf, Hagen, Delaney, Davis & Zinn (2020). The influence of social media on acne treatment: a cross-sectional survey. *Pediatric Dermatology*, 37(2), 301-304. [Baseline survey on social media influence on acne treatment decisions] <https://pubmed.ncbi.nlm.nih.gov/31944456/>
- AAD Acne Clinical Guidelines — primary ground-truth rubric for claim accuracy classification. <https://www.aad.org/member/clinical-quality/guidelines/acne>
- ACNE04 dataset — publicly available annotated acne image dataset for YOLOv8 severity scoring training. <https://www.kaggle.com/datasets/rutviklathiyateksun/acne04-grading-dataset>
- Apify TikTok Scraper — metadata collection tool (~\$5/run). <https://apify.com/clockworks/free-tiktok-scraper>
- pyktok — Python TikTok video download module. <https://github.com/dfreelon/pyktok>